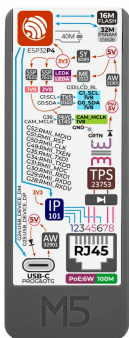
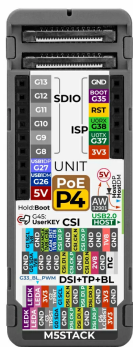


Unit PoE-P4

SKU:U213





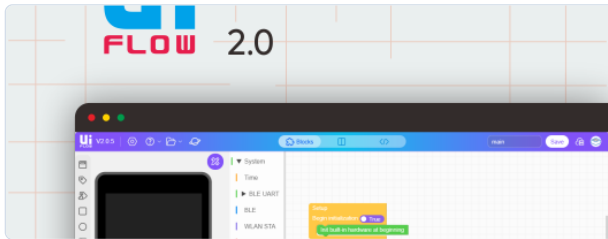
Description

Unit PoE-P4 is a high-performance PoE Ethernet embedded controller powered by the **ESP32-P4** SoC (RISC-V 32-bit dual-core @ 360MHz + LP single-core @ 40MHz). It integrates **16MB Flash** and **32MB PSRAM**, and features the built-in **IP101GRI** Ethernet PHY supporting **10/100 Mbps** communication. The device also supports PoE power delivery with a maximum output of 6W, and includes an onboard DC-DC converter to supply system power, greatly simplifying power wiring in practical applications.

It is equipped with a 24P FPC display expansion interface (MIPI DSI + TP + BL) and a 24P FPC camera interface (MIPI CSI), supporting high-resolution display output and image capture. For peripheral expansion, it provides one USB Type-C Host (USB 2.0) port for connecting U disks, HID devices, and other peripherals, plus one USB Type-C (USB 2.0) port supporting OTG and firmware download for debugging. In addition, it features a Hat expansion interface, SDIO & ISP IO expansion interfaces, and a Grove interface, enabling flexible connection to a wide range of sensors and peripherals.

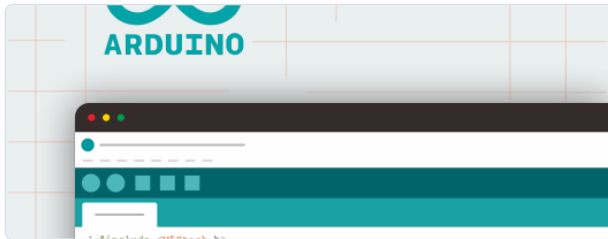
The device also integrates an RGB indicator LED, an infrared transmitter circuit, and a user button to support status indication, local interaction, and infrared signal transmission. It is ideal for applications such as industrial HMI, PoE smart terminals, vision acquisition nodes, smart home controllers, and edge computing devices.

Tutorial



UiFlow2

This tutorial introduces how to control Unit PoE-P4 through the UiFlow2 graphical programming platform.



Arduino IDE

This tutorial introduces how to program and control the Unit PoE-P4 using the Arduino IDE.

Features

- ESP32-P4 main controller:
 - 16MB Flash + 32MB PSRAM
- Peripheral interfaces:
 - 1x 24P FPC camera interface: MIPI CSI 2-Lane
 - 1x 24P FPC display expansion interface: MIPI DSI 2-Lane + TP + BL

- 1x USB Type-C Host interface (USB 2.0)
- 1x USB Type-C firmware download / OTG interface (USB 2.0)
- Ethernet:
 - Integrated IP101 10/100M Ethernet PHY chip
 - Supports PoE (active Ethernet) power supply, enabling data transmission and power delivery over a single Ethernet cable
- Human-machine interaction:
 - 1x RGB LED indicator
 - 1x user button
 - 1x infrared transmitter
- Expansion interfaces:
 - HY2.0-4P expansion interface
 - Hat2-Bus 2.54-16P
 - SDIO-Bus 2.54-9P
 - ISP-Bus 2.54-6P

| Includes

- 1 x Unit PoE-P4
- 1 x HY2.0-4P Grove Cable (20cm)

| Applications

- Industrial Human-Machine Interface (HMI)
- PoE smart terminals
- Vision acquisition nodes
- Smart home controllers
- Edge computing devices

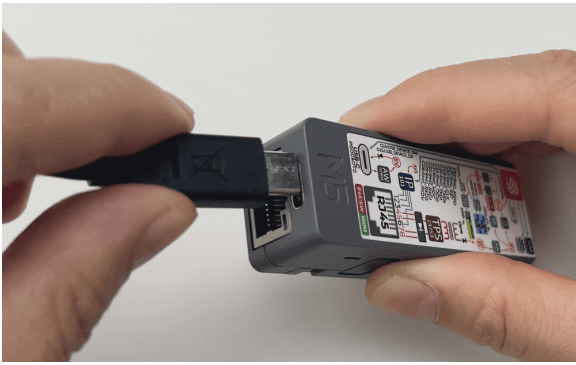
| Specifications

Specification	Parameter
SoC	ESP32-P4NRW32 (RISC-V 32-bit high-performance dual-core processor @ 360 MHz + RISC-V 32-bit low-power single-core coprocessor @ 40 MHz)
Flash	16MB
PSRAM	32MB Octal
Power Input	DC 5V (USB Type-C) / PoE
Ethernet Controller	IP101GRI
Ethernet Speed	10/100M adaptive
Ethernet Interface	RJ45
PoE Standard	Compliant with IEEE 802.3at
PoE Power	Maximum power 6W
Display Interface	24P FPC (0.5mm pin pitch) MIPI DSI 2-Lane + TP + BL
Display Resolution	MIPI DSI display supports up to 1920×1080 (limited by lane count and frame rate)
Camera Interface	24P FPC (0.5mm pin pitch) MIPI CSI 2-Lane
USB Type-C	1x USB Type-C Host interface (USB 2.0) 1x USB Type-C firmware download / OTG interface (USB 2.0)
Power Consumption	Operating current: 5V @ 73.82mA Infrared transmission: 3.3V @ 23.57mA (instantaneous) RGB: 5V @ 75.32mA Full load (display, camera, IR, RGB, etc.): 277.10mA Deep sleep: 5V @ 19.85mA
RGB LED	1 x NH-B2020RGBA-HF
Infrared Transmitter	1 x MHS153IRCT
Operating Temperature	0 ~ 40°C
Product Size	64.0 x 24.0 x 20.2mm
Product Weight	28.2g
Package Size	65.0 x 33.0 x 21.2mm
Gross Weight	35.0g

Learn

Download Mode

To flash firmware, connect the device to a computer via a USB Type-C data cable. Press and hold the side button for 3 seconds until the green LED lights up. The device will then enter download mode and wait for firmware flashing.



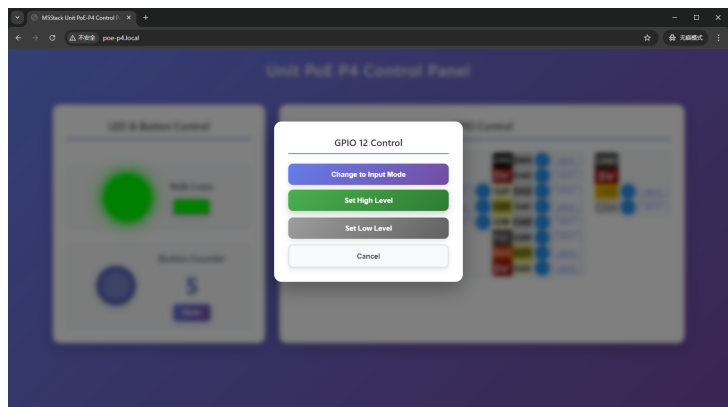
Access Device Control Page

The factory default firmware of the device is preloaded with a WEB control panel, which supports RGB LED color control, button press count statistics, GPIO pin mode switching, and high/low level output. The device control page can be accessed by following these steps:

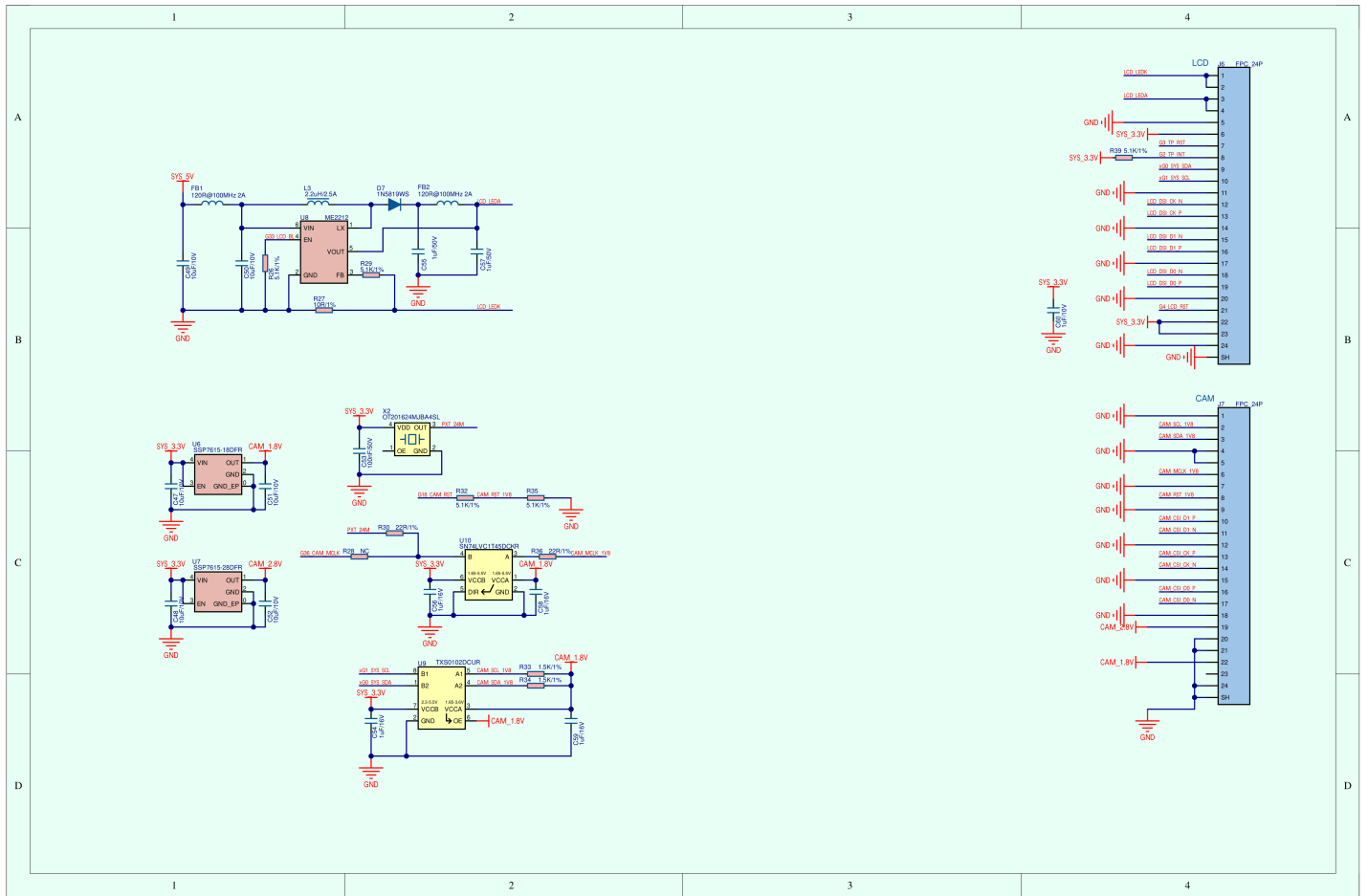
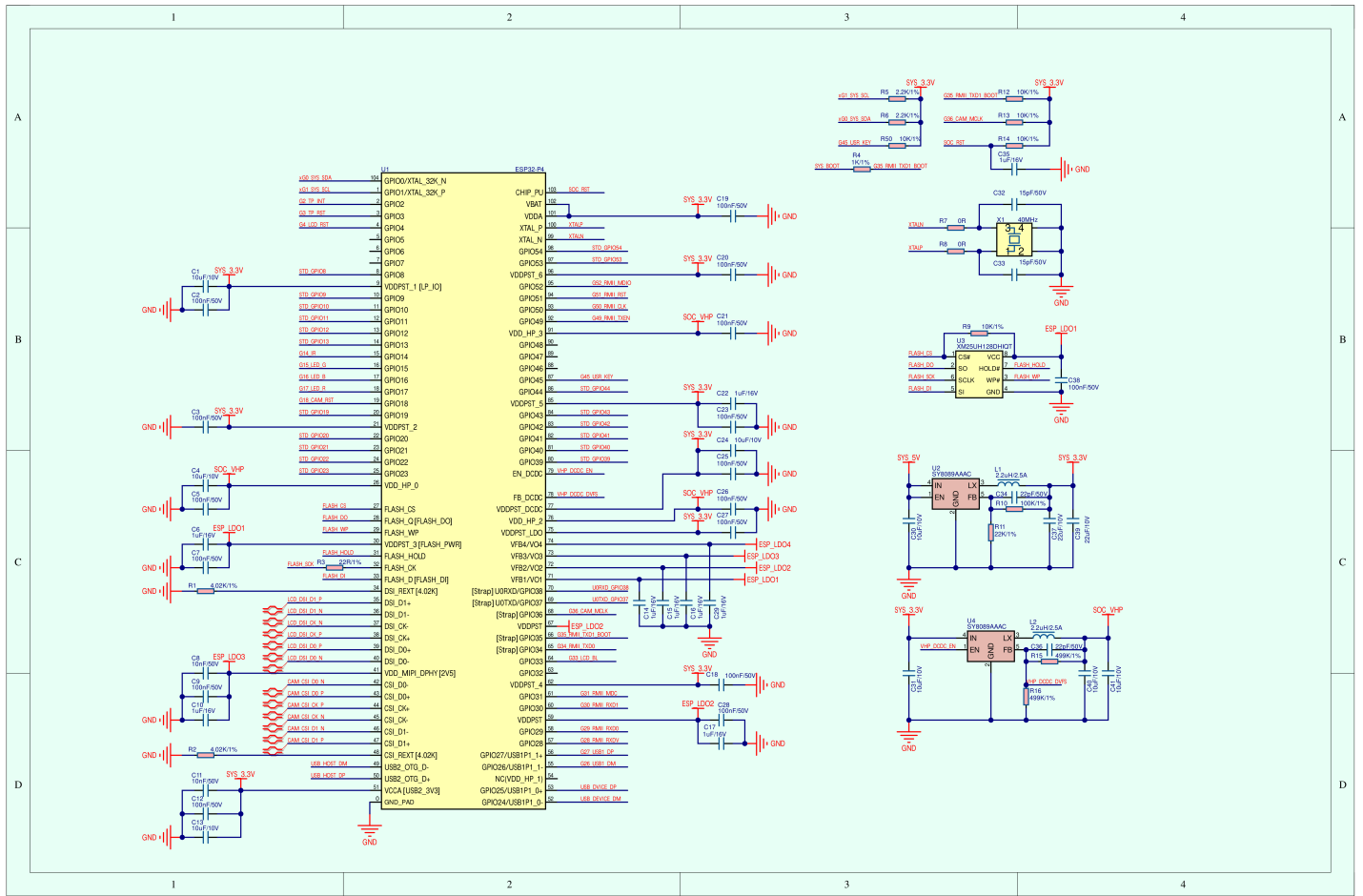
1. Power on the Unit PoE-P4 and connect it to the Ethernet network.
2. Open the serial monitor and note the device' s current IP address.
3. Ensure your computer or mobile phone is on the same local area network (LAN) as the device.
4. Access the web control page using one of the following methods:

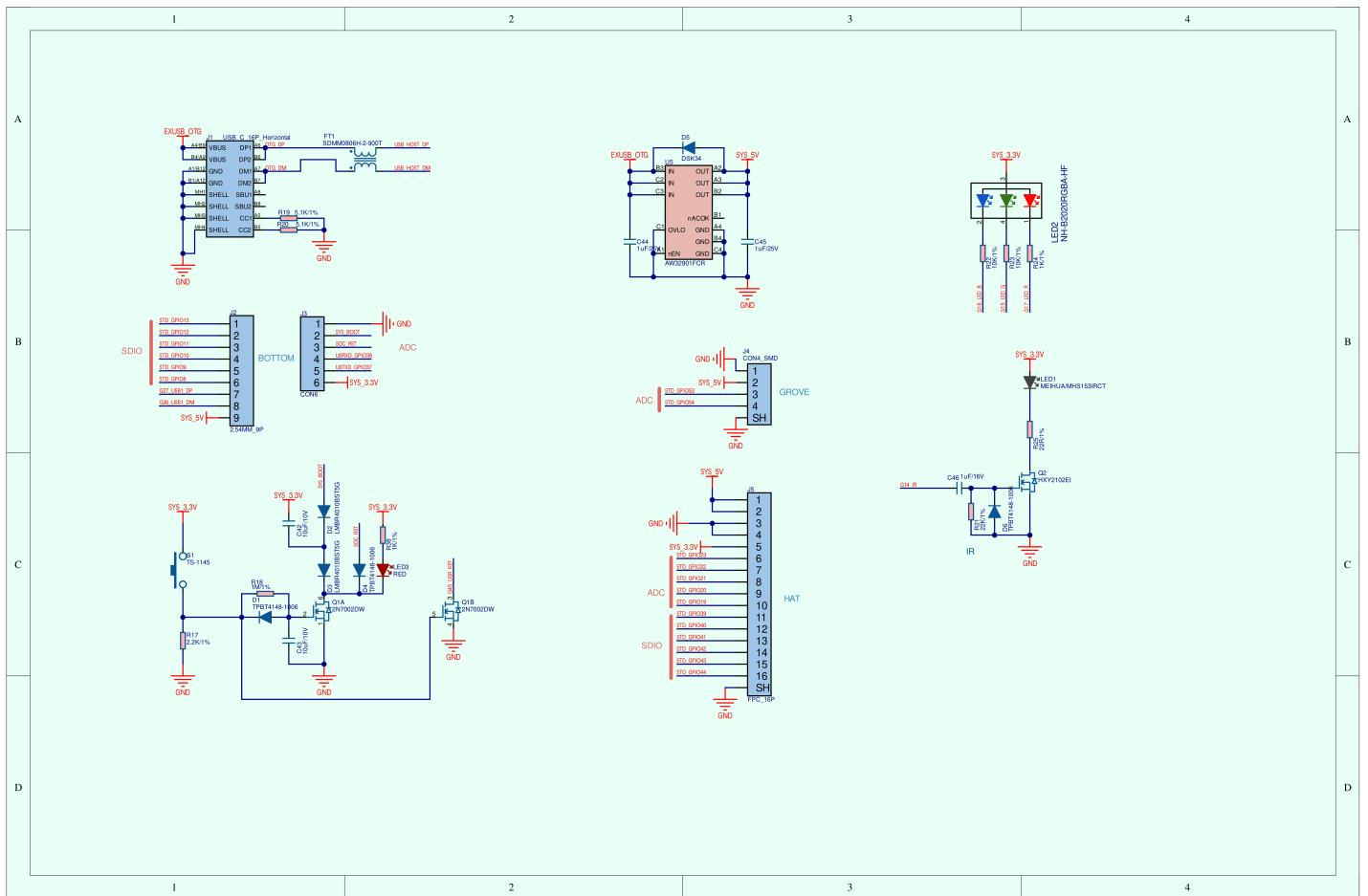
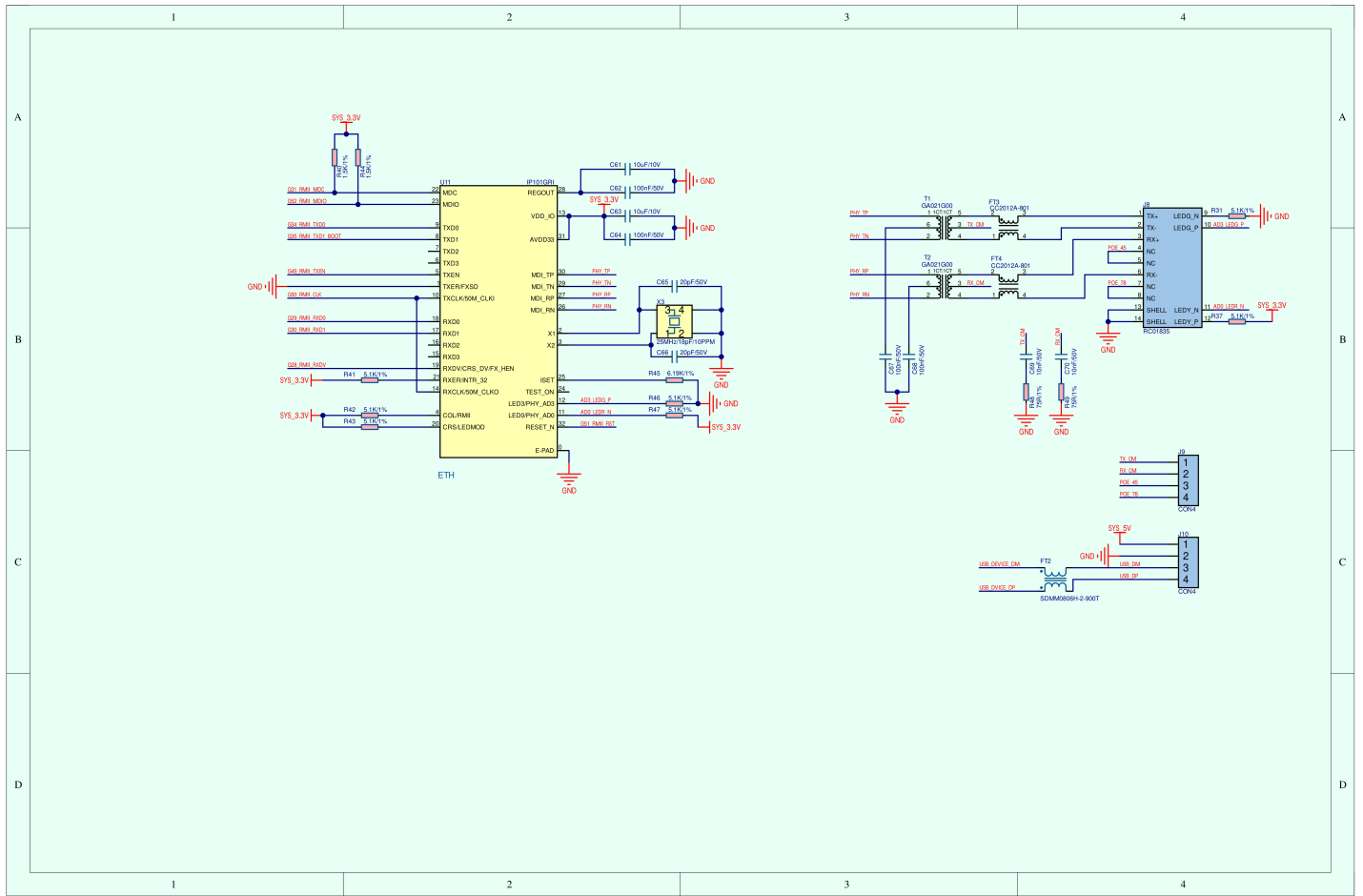


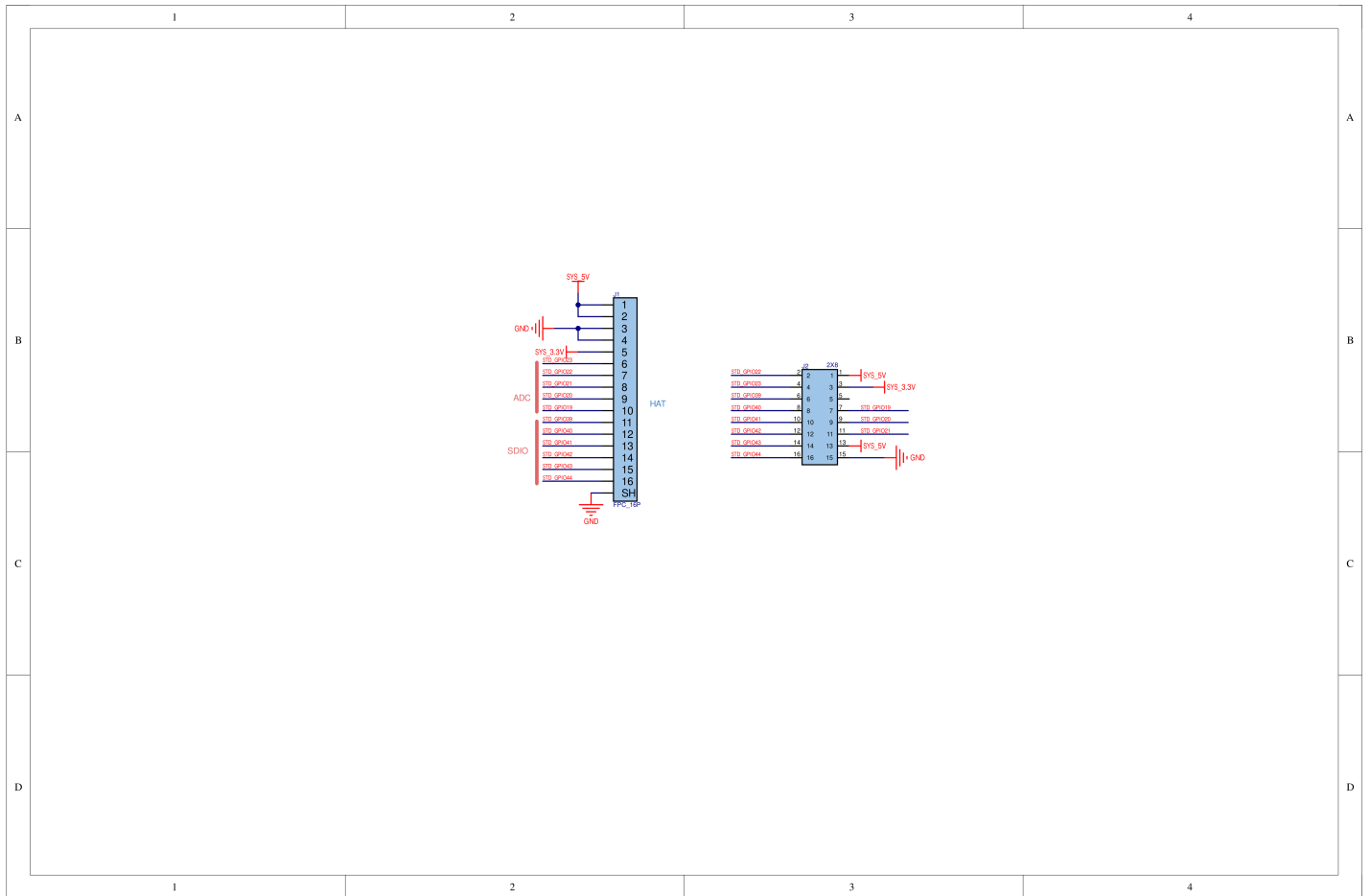
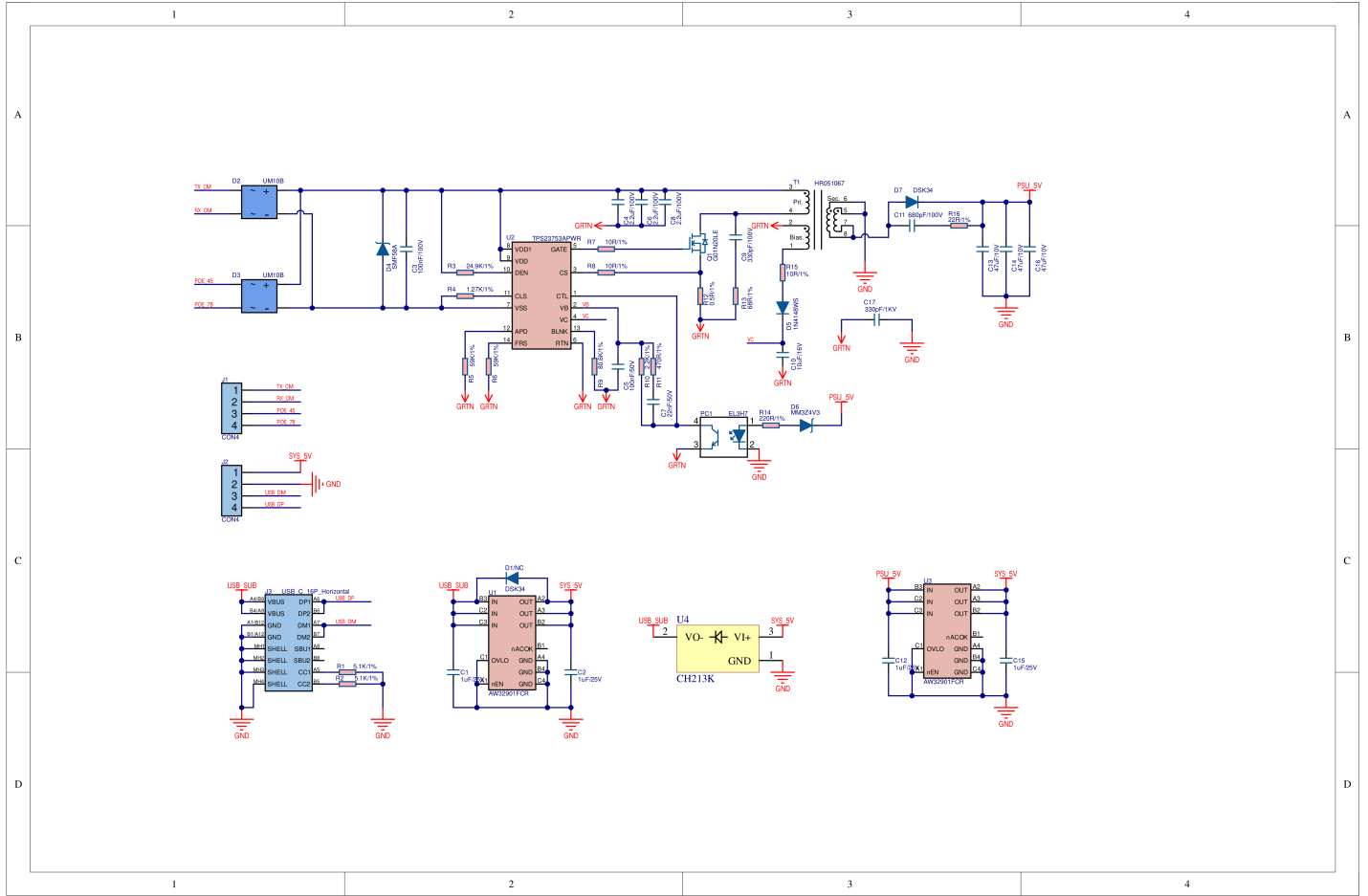
- Enter `poe-p4.local` in your browser
- Copy the IP address shown in the serial monitor and paste it into the browser address bar



- o [Unit PoE-P4 Mainboard Schematics PDF](#)
- o [Unit PoE-P4 Power Board Schematics PDF](#)
- o [Unit PoE-P4 HAT Expansion Interface Schematics PDF](#)









PinMap

Ethernet

ESP32-P4	G31	G52	G51
IP101	SMI_MDC	SMI_MDIO	PHY_RST

ESP32-P4	G28	G29	G30	G34	G35	G49	G50
IP101	RMII_RXDV	RMII_RXD0	RMII_RXD1	RMII_TXD0	RMII_TXD1	RMII_TXEN	RMII_CLK

LCD

ESP32-P4	G4	G33	DSI_CLK_N	DSI_CLK_P	DSI_D1_N	DSI_D1_P	DSI_D0_N	DSI_D0_P
FPC - MIPI DSI	LCD_RST	LCD_BL	LCD_DSI_CK_N	LCD_DSI_CK_P	LCD_DSI_D1_N	LCD_DSI_D1_P	LCD_DSI_D0_N	LCD_DSI_D0_P

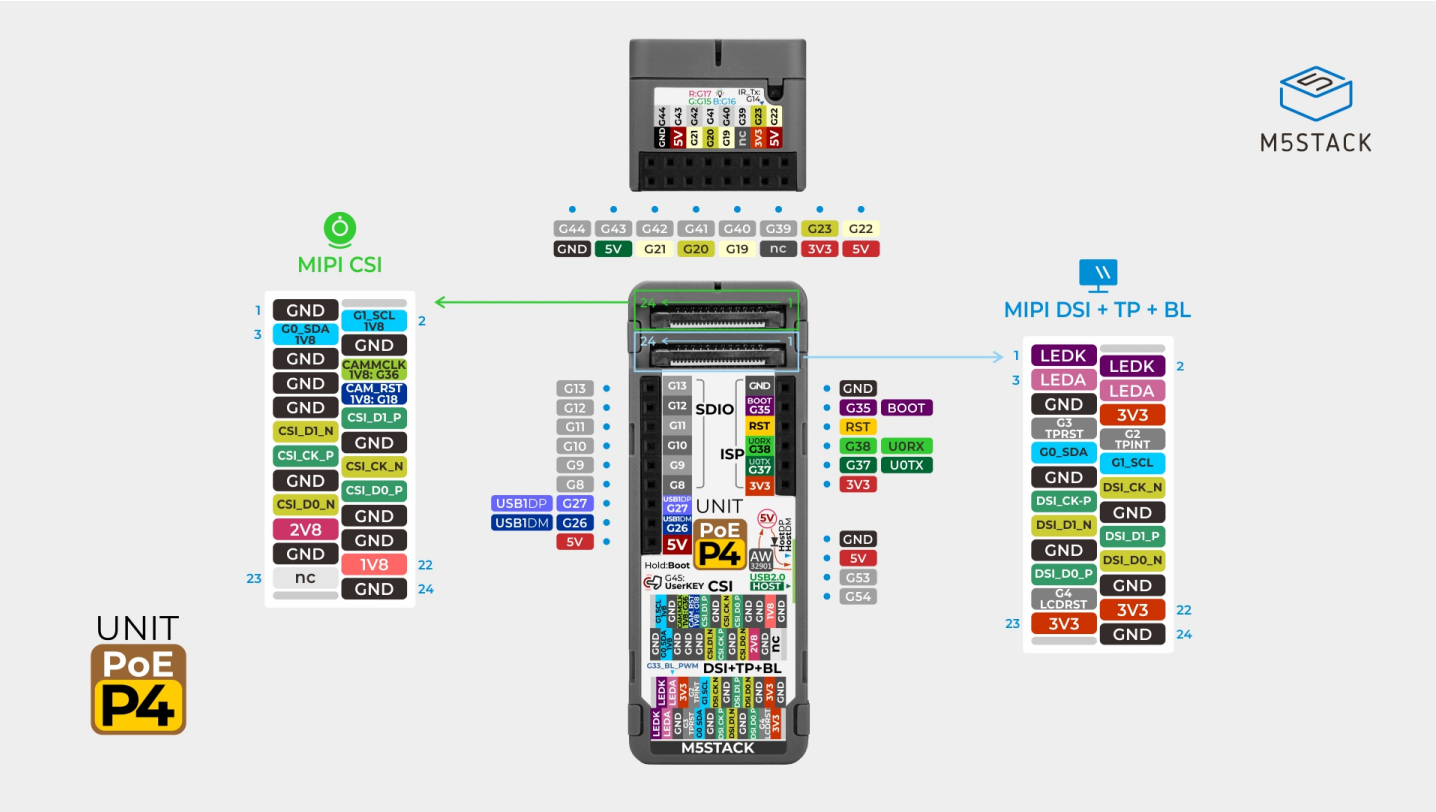
TP

ESP32-P4	G0	G1	G3	G2
FPC - TP	SDA	SCL	TP_RST	TP_INT

Camera

ESP32-P4	CSI_CK_N	CSI_CK_P	CSI_D1_N	CSI_D1_P	CSI_D0_N	CSI_D0_P
MIPI CSI FPC	CAM_CSI_CK_N	CAM_CSI_CK_P	CAM_CSI_D1_N	CAM_CSI_D1_P	CAM_CSI_D0_N	CAM_CSI_D0_P

ESP32-P4	G0	G1	G18	G36
MIPI CSI FPC	SDA_1V8	SCL_1V8	CAM_RST	CAM_MCLK



USB Type-C

ESP32-P4	USB2_OTG_DP	USB2_OTG_DN
USB 2.0 HOST	HOST_DP	HOST_DN

ESP32-P4	G25	G24
USB 2.0 Download / OTG	USB0_DP	USB0_DN

IR & RGB LED & Button

ESP32-P4	G14	G15	G16	G17	G45
IR	IR_TX				
RGB LED		LED_GREEN	LED_BLUE	LED_RED	
Button					KEY_USER

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G53	G54

SDIO & ISP

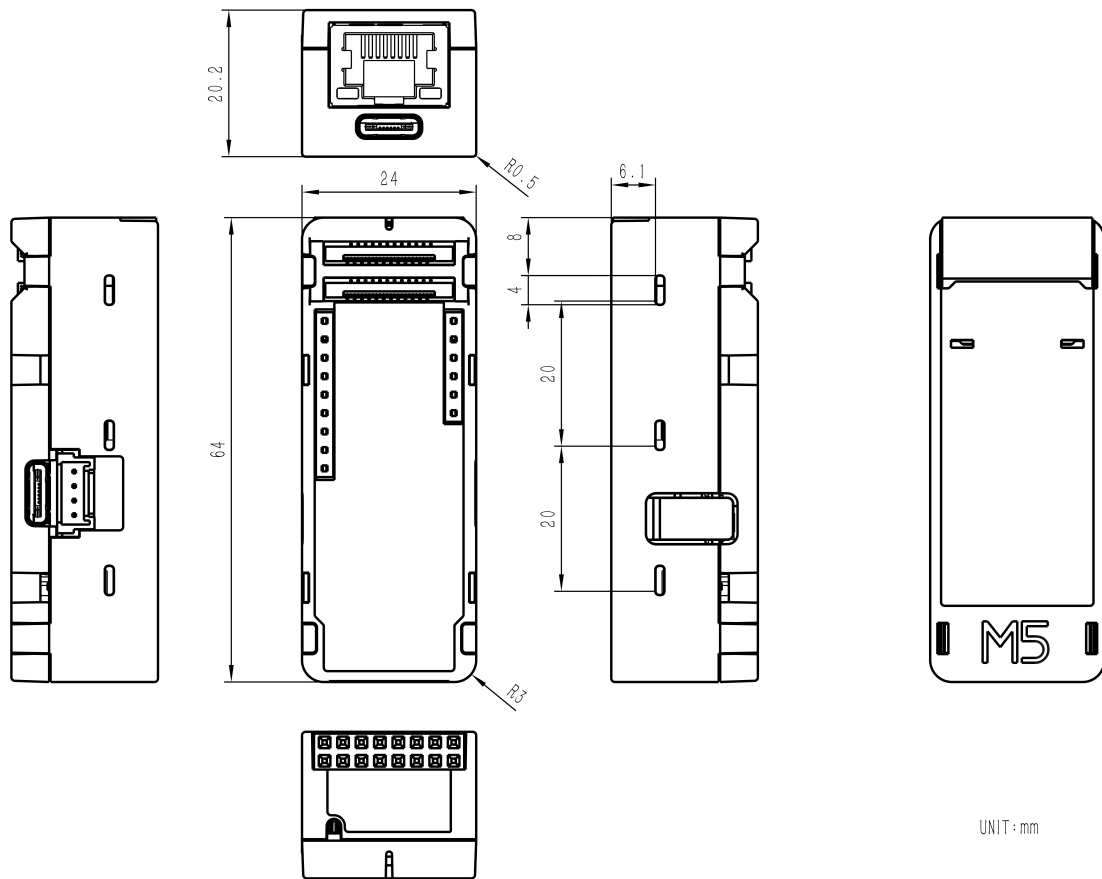
FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	G13	1	1	GND	
	G12	2	2	G35	BOOT
	G11	3	3	RST	
	G10	4	4	G38	UART0_RX
	G9	5	5	G37	UART0_TX
	G8	6	6	3V3	
USB1_DP	G27	7			
USB1_DM	G26	8			
	5V	9			

Hat2-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	G44
5V	3	4	G43
G21	5	6	G42
G20	7	8	G41
G19	9	10	G40
NC	11	12	G39
3V3	13	14	G23
5V	15	16	G22

Model Size

- Unit PoE-P4 Model Size PDF



Structure

- [Unit PoE-P4 Structure Files](#)

Datasheets

- [ESP32-P4](#)
- [IP101GRI](#)
- [TPS23753A](#)

Softwares

Arduino

- [Unit PoE-P4 Arduino Quick Start](#)

UiFlow2

- [Unit PoE-P4 UiFlow2 Quick Start](#)

ESP-IDF

- [Unit PoE-P4 Factory Firmware Source Code](#)
- [Unit PoE-P4 Factory Firmware Build Guide](#)

Other

- [Unit PoE-P4 Factory Reset Firmware Tutorial](#)

| Video
